BeatAI: Guided Beat Generator for Beginners

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What We Needed to Validate

- Beginners prefer guided structure instead of full automation
- Users want to edit specific sections (not regenerate the entire song)
- AI explanations of music theory would actually help learning
- Connecting lyrics + beat structure is more useful than separate tools

Tools We Tested

1. ChatGPT

Pros:

- Strong structured outlines
- Clear BPM + chord explanations

Cons:

- No real audio output
- No rhythm-to-lyric alignment
- No direct exportable MIDI

2. Gemini

Pros:

- Structured plans
- Beat/tempo analysis
- Playable music

Cons

- Lyrics random or unnatural
- Educational explanations only appear if heavily prompted

Tools We Tested

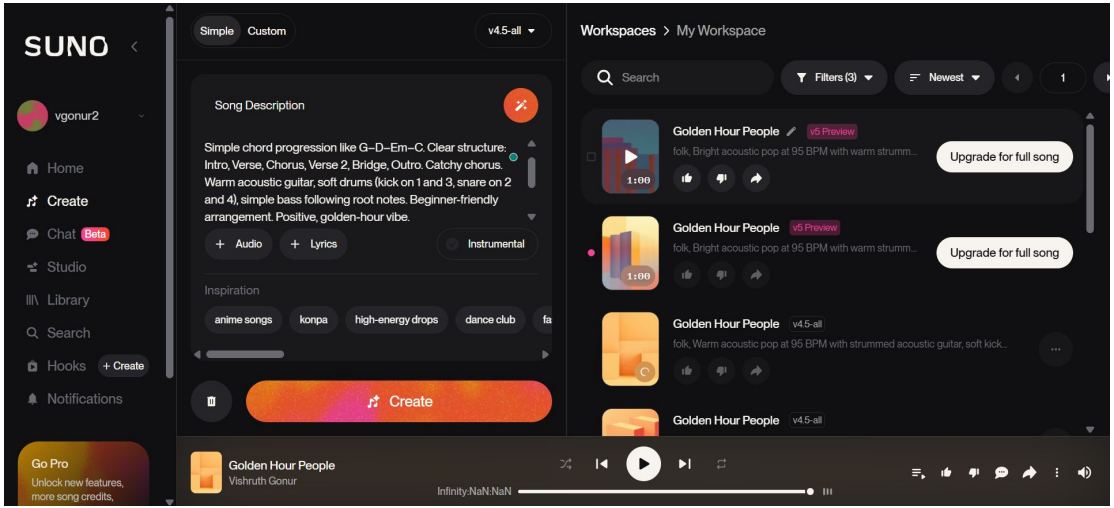
3. Suno

Pros:

- High-quality full songs instantly
- Clear mood + vibe matching

Cons

- Sad vs. happy songs sounded very similar
- No transparency into chord/drum logic



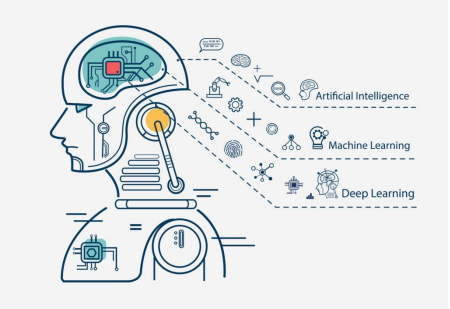
What Worked

- AI tools generate ideas extremely fast
- ChatGPT provides clearly labeled song sections
- Suno created high-quality, polished audio
- AI is good for inspiration and overcoming creative block

What Didn’t Work

	Emotional Difference Problem	Lyrics Quality + Assessment	Educational Gap
Vulnerabilities	Suno outputs are consistent across moods	Gemini and other tools produced random lyrics	Lack of teaching for beginners
What we want to fix	Clearer emotional mapping	Beats only? or beats + lyric alignment?	Education built into responses and UI

How Our Concept Changed



After Prompt Testing, we realized:

- Emotional changes weren’t reflected in chords or BPM
- Full-song generators remove user control
- Text-only tools don’t align with musical structure

In our design, we:

- Prioritized BPM + chord alignment based on mood
- Emphasize MIDI export instead of full MP3 generation
- Kept a simple and concise UI

Risks & CP3 Plan

Risks:

- Only runs locally
- Incorrect or hallucinated music theory explanations
- Over-reliance on AI reducing creativity

For CP3 we plan to:

- Explore sample music datasets on Hugging Face to understand available MIDI and music structure datasets for prototyping; Right now, we have mock dink
- Implement editable sections in our v0.dev prototype
- Add explanation toggle and Add basic MIDI export functionality

Thank You!